CLINE AI BUILDER HACKATHON · ROUND 2

Nyaya-Ledger

Nyaya (न्याय / ನ್ಯಾಯ) means justice.

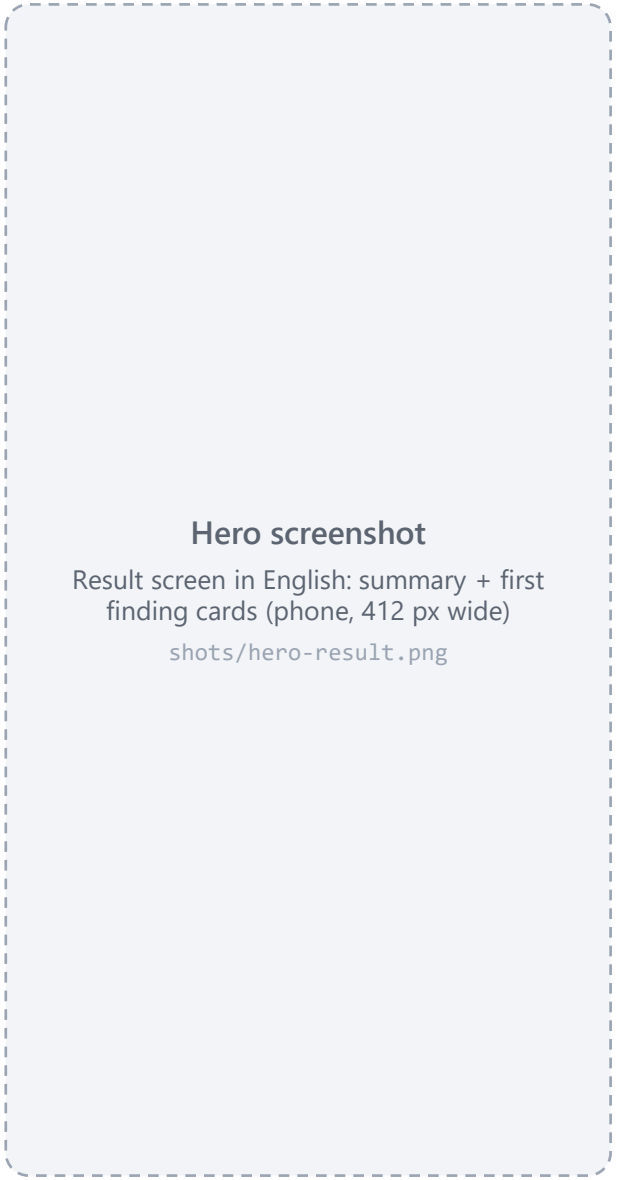
Turns the payout screenshots and bank SMS a gig worker already keeps into a checked, law-grounded grievance docket.

Cline SDK agent inside | Karnataka Gig Workers Act, 2025 | Kannada · Hindi · English

Devansh Jangid · Debasis Patnaik

Live nyaya-ledger.onrender.com

Code github.com/devanshjangid2005/nyaya-ledger



02 THE PROBLEM

Paid, penalised and blocked by systems they can't inspect

7.7M → 23.5M

gig workers in India, 2020-21 → projected 2029-30 (NITI Aayog)

~7 lakh

active gig workers reported by platforms in Karnataka, April 2026

83% · 87%

of cab drivers · delivery workers say ID blocking or deactivation hurt them (PAIGAM 2024)

WHAT A WORKER ACTUALLY SEES

`Adjustment  −₹180`

A deduction with no reason given

`App ₹6,105 · Bank ₹6,005`

Less money arrives than the app shows

`"Your ID has been blocked"`

No reason, effective the same day

The platform holds the data. The worker holds a pile of screenshots.

Sources: NITI Aayog (2022); The Hindu (2026); PAIGAM, Prisoners on Wheels? (2024). Links in the README.

03 WHY NOW

The rights and the filing channel exist. The evidence doesn't.

KARNATAKA PLATFORM BASED GIG WORKERS ACT, 2025

s.15(1)	Reasons for every deduction, stated in the invoice; pay as per contract
s.15(2)	Pay at least weekly, with no delay
s.14(2)	No deactivation without written reasons and 14 days' notice
s.13(1)	Right to ask how automated systems affect earnings
s.22(2)	Grievance on payouts, deductions or termination to the platform's IDRC

TIMELINE

30 May 2025
Act in force

1 May 2026
Grievances filed via IPGRS, routed to the IDRC

3 Jul 2026
High Court declines to stay the Act; case pending

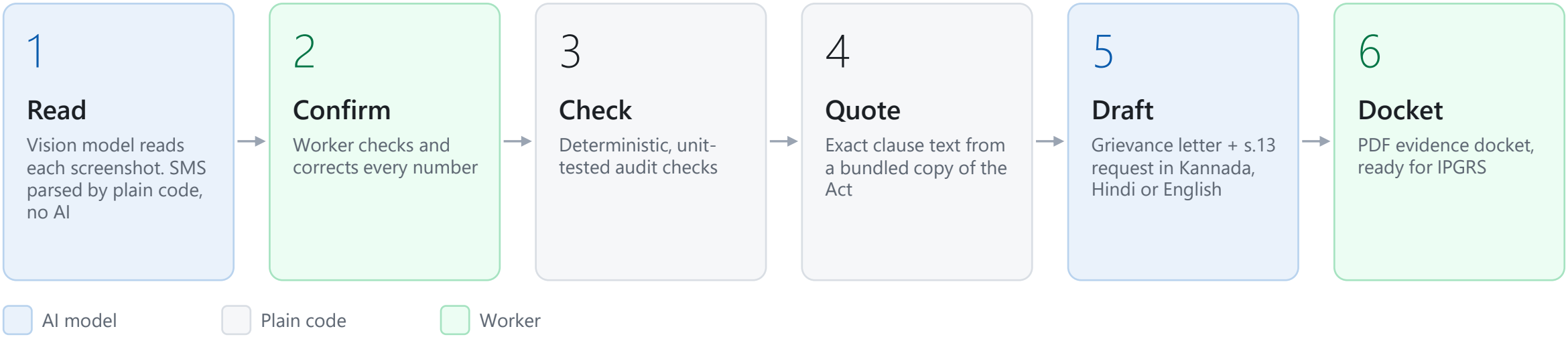
Missing: the evidence

It sits in the worker's gallery and SMS inbox, and nobody reconciles it by hand.

04 THE SOLUTION

From a pile of screenshots to a docket a committee can act on

Upload payout screenshots, paste bank SMS, add hours and fuel cost. Nyaya-Ledger does the rest:



WHAT IT DOES NOT CLAIM

Not a verdict

Findings say "possible issue for review", with the arithmetic shown

Not legal advice

Every docket says so, and notes the pending High Court case

No platform login

No partner API exists; scraping could get the worker blocked

One week of evidence, audited in about 30 seconds

Collect

Collect screen: upload, SMS box, "Try with demo data"

`shots/shot-collect.png`

1 Collect

Confirm

Confirm screen: thumbnails + editable numbers + bank credits

`shots/shot-confirm.png`

2 Confirm

Docket

Result screen: finding cards with arithmetic

`shots/shot-result.png`

3 Docket

WHAT IT FOUND (SYNTHETIC DATA)

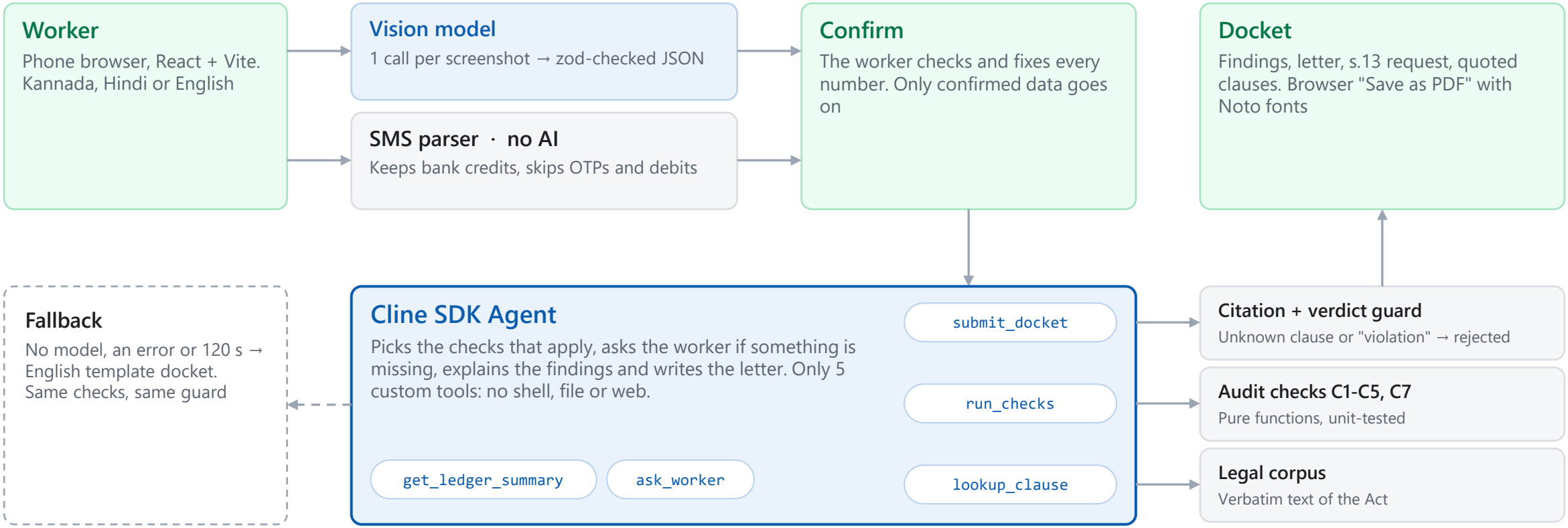
C1	**₹275**	of deductions with only a vague label
C2	**₹120**	gap: ₹6,650 − ₹425 ≠ ₹6,105 shown
C3	**0 days**	notice and no reason for an ID block
C7	**₹100**	less reached the bank than the app showed
C5	**₹90/h**	real earnings after fuel, against ₹128/h gross

C4 OK weekly payouts. OTP and fuel-debit SMS skipped automatically.

Output: 5 possible issues, grievance letter, s.13 request, PDF docket.

06 ARCHITECTURE

AI reads and writes. Code decides the numbers.



STACK

TypeScript · Node 22 | Express + multer | React + Vite | @cline/sdk | Gemini 3.1 Flash-Lite | zod | vitest · 42 tests | Render free tier

AI where language is hard. Code where numbers must be right.

AI DOES

Read screenshots

Gemini 3.1 Flash-Lite (vision) → strict JSON, validated with zod, 1 retry. Every value keeps its source image ID

Run the audit

Cline SDK Agent on the same model: decides which checks apply, asks the worker for missing inputs

Explain and draft

Plain-language summary, grievance letter and s.13 request in Kannada, Hindi or English

CODE DOES (NO AI)

Parse bank SMS

Regex parser for SBI, HDFC and Axis-style credits. SMS never reach a model

Every number

Checks C1-C5 and C7 are pure TypeScript functions with vitest tests

Every citation

Clause text comes from a bundled copy of the Act, never from the model

THE AGENT'S ONLY 5 TOOLS

`get_ledger_summary`	`run_checks`	`lookup_clause`	`ask_worker`	`submit_docket`
what's in the ledger	deterministic checks	verbatim Act text	one question, live	guarded, ends the run

Cline built it. The Cline SDK runs inside it.

IN THE BUILD

Cline in the build

A3 clip frame: Cline panel with the failing SMS test → the fix in sms.ts → tests green

shots/cline-build.png

Test-first Failing vitest test first, then Cline writes the check until it passes

Built with Cline Scaffold, zod schemas, citation guard, React screens, e2e test

42 tests checks, SMS parser, guards, agent loop with a stub model, API

INSIDE THE PRODUCT · BONUS CRITERION

```
const agent = createAgent(source, {
  systemPrompt: systemPrompt(ledger),
  tools: [getLedgerSummary, runChecks,
          lookup, ask, submit],
  maxIterations: 12,
});

createTool({ name: "submit_docket",
  lifecycle: { completesRun: true }, ... })
```

server/agent.ts

Agent + createTool 5 zod-typed custom tools, no built-in shell, file or web tools

ask_worker Pauses the run and asks the worker a question over SSE

completesRun submit_docket ends the run, only after the guards pass

09 CHALLENGES

What could go wrong, and what we built for it

CHALLENGE		WHAT WE DID
Vision misreads a number	→	The worker confirms every value before any check runs
Bank SMS formats vary	→	Test-first parser; unknown formats are skipped with a reason, credits can be added by hand
Prompt injection in screenshots	→	Agent has only 5 typed tools. Ledger text is treated as data, never as instructions
Model invents a law or a verdict	→	Citation guard + verdict-word guard (EN · ಕನ್ನಡ · हिन्दी) reject the docket; agent must rewrite
Kannada PDF on a 512 MB free host	→	No server Chromium: print stylesheet + Noto fonts, the phone's browser saves the PDF
Free-tier quota and timeouts	→	Template fallback after 120 s, same checks and guard, labelled "generated without AI"
The Act is challenged in court	→	No position taken; every docket carries the pending-case note

Real example: in testing, Gemini wrote ಉಲ್ಲಂಘನೆ (violation) in a Kannada letter. The verdict guard caught it and the agent rewrote it.

Karnataka today. Every state with a gig-worker law next.

LIMITATIONS

- Karnataka only
- Accuracy depends on screenshot quality
- Can't see platform-side data: incentive rules, welfare-fee deposits
- Not legal advice; the Act's validity is pending in the High Court

FUTURE SCOPE

More states Rajasthan, Telangana, Bihar, and the central Code on Social Security

Less typing Voice input in Kannada and Hindi; screen-recording and bank-statement import

Android companion Reads bank SMS directly (Play Store money-management exception)

Union dashboard Anonymised, aggregated findings across workers

Platform data DPDP access requests from May 2027; direct IPGRS filing if an API opens

Try it on your phone

Tap "Try with demo data"

nyaya-ledger.onrender.com

github.com/devanshjangid2005/nyaya-ledger

First load may take a minute (free server wakes up)